Jérôme Lin

Software Engineering Student — Seeking a 4-to-6-Month DevOps / Systems Internship

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EDUCATION

EPITA — School of Computer Science & Advanced Technologies Paris, France
Master of Science in Computer Science & Engineering (Expected Grad.: 2029) **2024 – Present**

- **Core Curriculum (ING1):** Operating Systems internals, C/UNIX programming, POSIX standards, Networking (TCP/IP stack, routing), Compilers, Database systems (SQL), Concurrency.
- **Academic background:** International Preparatory Cycle (taught in English); High School Baccalaureate in Mathematics & Physics-Chemistry (Mention Très Bien).
- **Erasmus+ Grant:** Eligible for EU mobility funding (Germany, Luxembourg, Netherlands).

TECHNICAL SKILLS

Systems & OS: Linux/UNIX internals, POSIX API, Process Management, Signals, Memory Management
Languages: C, C++, Python, POSIX Shell (Bash), C#, SQL, HTML/CSS, JavaScript/React
DevOps & Tools: Git, Docker, Make / Makefiles, GCC/Clang, GDB, Valgrind, GitHub Actions, Linux CLI
Networking & DB: TCP/IP, OSI model, Sockets, Firebase Realtime Database, JSON
Languages (Spoken): French (Native), English (B2/Fluent, English track), German (A2)

KEY TECHNICAL PROJECTS

POSIX Shell Implementation (42sh) | *C, POSIX API, Linux, AST, Git* **EPITA SAÉ**

- Architected a compliant POSIX shell from scratch in C under rigorous coding norms (-Wall -Wextra -Werror).
- Implemented process lifecycle control (`fork`, `execvp`, `waitpid`), I/O redirection pipelines (`pipe`, `dup2`), and asynchronous signal handling (`SIGINT`, `SIGTERM`).
- Built a robust lexical analyzer and recursive descent parser generating an Abstract Syntax Tree (AST).

Dynamic Memory Allocator (libmalloc) | *C, Linux Kernel API, mmap, Pthreads* **EPITA**

- Developed an overridden dynamic allocator library (`malloc`, `free`, `calloc`, `realloc`) using `mmap(2)` and `munmap(2)`.
- Optimized chunk metadata layout and implemented thread-safety via POSIX mutexes to prevent memory corruption.
- Integrated into host binaries via `LD_PRELOAD` to handle heavy multi-threaded memory workloads.

OCR & Word Search Solver | *C, Image Processing, Neural Networks* **Academic Project**

- Implemented an end-to-end character recognition pipeline in C without external high-level ML frameworks.
- Developed image pre-processing algorithms (noise reduction, binarization, grid line detection) and a custom multilayer perceptron for optical letter classification.

Minimake — Simplified Build System | *C99, Makefile Parsing, Process Scheduling* **EPITA**

- Built a UNIX `make(1)` clone executing dependency graphs, pattern rules (%), and variable expansions.
- Utilized timestamp comparison (`stat(2)`) to determine outdated targets and run parallel child subshells.

PROFESSIONAL EXPERIENCE

Saipem SA Montigny-le-Bretonneux, France
Engineering Intern — Drilling Machine Digital Twin **June 2025 – July 2025**

- Developed a digital twin application representing an industrial drilling rig for real-time visualization.
- Gained direct experience with corporate software lifecycle, cross-functional collaboration, and industrial safety constraints.

EPITA School Ambassador Paris, France
Student Representative & Presenter **2024 – Present**

- Represent the institution during academic forums, open days, and student onboarding sessions.
- Deliver technical demonstrations and guide prospective international students.